

Learned exchange-correlation functionals

Training-subset generation, pre-training, in-sample optimization and held-out generalization

2026-09-09

Scope

One network pair (exchange and correlation) per architecture and training-subset size, as separate cluster runs under one protocol, drawn once over every finished cell (the runs merged into one view). Each run carries a pre-training (the cloning of the parent), a fidelity certificate that gates its training array, and the array itself, one cell per (architecture, subset size).

Run	Architectures	Cells done / planned	Run directory
size	deep_3x16, deep_attn_3x16	deep_3x16 11/11, deep_attn_3x16 9/11	run_20260902T145245Z
families	deep0_3x16, deep0_cusp_3x16	deep0_attn_3x16, deep0_3x16 11/11, deep0_attn_3x16 2/11, deep0_cusp_3x16 0/11	run_20260902T145247Z
meta-GGA families	deep0_cusp_mgga_3x16, deep0_mgga_3x16, deep0_mgga_attn_3x16, deep0_rung35_mgga_3x16, deep0_rung35ms_mgga_3x16	0/55 (first certificate FAIL, four pending)	run_20260902T145250Z

- Subset sizes 1, 2, 3, 4, 5, 6, 7, 12, 15, 18, 26 points of the 26-point DFS pool [DFS21, SI Sec. II], chosen by Jensen-Shannon selection (Sec. 2).
- Basis 6-311++G(3df,2pd), grid level 3, density fitting, at every stage [cfg]. The validation-best checkpoint is read for every optimization figure.

1. Model: density variables and the DFS coordinates

With $n = n_\alpha + n_\beta$ and $\sigma = |\nabla n|^2$ [DFS21 Eqs. 4, 5]:

$$k_F = (3\pi^2 n)^{1/3}, \quad s = \frac{|\nabla n|}{2k_F n}, \quad r_s = \left(\frac{3}{4\pi n}\right)^{1/3},$$

$$\zeta = \frac{n_\alpha - n_\beta}{n}, \quad \phi(\zeta) = \frac{1}{2} \left[(1 + \zeta)^{4/3} + (1 - \zeta)^{4/3} \right].$$

The networks read logarithmic compressions of these [DFS21 Eqs. 7 to 10; code: networks.py:621-657]:

$$x_0 = \ln(n^{1/3} + 10^{-5}), \quad x_1 = \ln \phi(\zeta), \quad x_s = (1 - e^{-s^2}) \ln(s + 1), \quad x_\alpha = \ln \frac{\alpha + 1}{2}.$$

- Exchange input $[x_s, \dots]$ with s of the doubled spin channel $2n_\sigma$ (the exchange network is evaluated per spin channel, next slides); correlation input $[x_0, x_1, x_s, \dots]$ on the total density. The ellipsis holds the architecture's extra descriptors; on the meta-GGA rung the raw α is replaced by x_α .
- The 10^{-5} in x_0 is the DFS reference implementation's constant [dpyscf net.py:39; code: networks.py:27].

1. Model: the extra descriptors

Iso-orbital indicator (meta-GGA rung; SCAN's α [SCAN15 Eq. 2; DFS21 Eq. 6; code: metagga.py:8–9]), a contraction of the live density matrix P against the AO gradients, hence self-consistent and differentiable through the SCF:

$$\alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}}, \quad \tau = \frac{1}{2} \sum_i |\nabla \psi_i|^2 = \frac{1}{2} \sum_{\mu\nu} P_{\mu\nu} \nabla \chi_\mu \cdot \nabla \chi_\nu,$$

$$\tau_W = \frac{|\nabla n|^2}{8n}, \quad \tau_{\text{unif}} = \frac{3}{10} (3\pi^2)^{2/3} n^{5/3},$$

with $\alpha \geq 0$ imposed by a smooth positive part rather than a hard clip [code: metagga.py:25–31].

Nuclear cusp pair (cusp), two bounded geometric features per grid point [code: descriptors.py:230–245]:

$$x_4 = \exp(-2Z_{\text{nearest}} r_{\min}), \quad x_5 = \tanh\left(\frac{1}{5} \ln \sum_A \frac{Z_A}{r_A}\right),$$

x_4 the density-form cusp envelope e^{-2Zr} [Steiner63, after Kato57], x_5 the log-compressed bare-nuclear potential. Both are geometric, so the synthetic pre-training mesh (Sec. 3) carries no values for them.

1. Model: the functional form

MLP output g at a grid point (GELU, depth $\times$ nodes) [code: networks.py:169, 379–404]:

$$F_x = 1 + \mathcal{L}_{1.804}(G_x \cdot g_x), \quad F_c = 1 + \mathcal{L}_2(G_c \cdot g_c),$$

$$\mathcal{L}_\lambda(z) = \lambda \operatorname{sigmoid}(z - \ln(\lambda - 1)) - 1 \quad [\text{DFS21 Eq. 11}],$$

$$G = \tanh^2(s) \text{ (GGA)}, \quad G = x_s + \tanh^2(x_\alpha) \text{ (meta-GGA) [DFS21 Eq. 12].}$$

$\mathcal{L}_\lambda(0) = 0$ and $F \in (0, \lambda)$; $\lambda = 1.804 = 1 + \kappa_{\text{PBE}}$ for exchange [PBE96 Eq. 14] (DFS uses 1.174 [PRSB14]), $\lambda = 2$ so that $F_c \geq 0$ [DFS21 Eq. 13]. The gate G pins $F = 1$ where the gradient vanishes.

$$E_{xc} = \frac{1}{2} (E_x[2n_\alpha] + E_x[2n_\beta]) + E_c[n_\alpha, n_\beta] \quad [\text{OP79; code: descriptors.py:124}],$$

$$E_x[n] = \int n \epsilon_x^{\text{LDA}}(n) F_x d^3r, \quad E_c = \int n \epsilon_c^{\text{PW92}}(r_s, \zeta) F_c d^3r \quad [\text{PW92}].$$

Attention variants (`_attn`) [Vas17 Eq. 1; Xio20; code: net.py:55–136]: after the first hidden layer, the H hidden units are h tokens of dimension $d_k = H/h$ under pre-layer-norm self-attention across the token axis, a per-point re-mixing of hidden units:

$$\text{Attn}(Q, K, V) = \operatorname{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad x \leftarrow x + W^O \operatorname{Concat}(\text{head}_1, \dots, \text{head}_h)(\text{LN}(x)).$$

1. Model: the architectures

Architecture	Last layer	Rung, parent	Hidden	Heads	Extra
deep_3x16	Glorot	GGA, PBE	3 x 16	none	none
deep_attn_3x16	Glorot	GGA, PBE	3 x 16	4	none
deep0_3x16	zeroed	GGA, PBE	3 x 16	none	none
deep0_attn_3x16	zeroed	GGA, PBE	3 x 16	4	none
deep0_cusp_3x16	zeroed	GGA, PBE	3 x 16	none	cusp
deep0_cusp_mgga_3x16	zeroed	meta-GGA, SCAN	3 x 16	none	cusp, α

- Inputs: F_x from x_s , F_c from x_0, x_1, x_s ; the cusp architectures add x_4, x_5 to both, the meta-GGA one x_α as well.
- deep_3x16 and deep0_3x16 have the same layers, inputs and parameter count; deep0_3x16 differs only in a zero-initialized final layer, $F_x = F_c = 1$ at initialization. After pre-training both are PBE clones differing only in their fitted weights: the two columns are replicates of one architecture under two starts; their combined held-out verdicts agree at every subset size, and on the energy leg alone they differ at 3 and 4 points (Sec. 5).

2. Subsets: the 26-point DFS pool [DFS21 SI Sec. II]

21 atomization energies (G2/97 molecules), 3 BH76 reaction energies, 2 IP13 ionization potentials; each point carries the species its loss term needs (an atomization its molecule and the H and Li anchors it contains, a reaction every reactant and product, an ionization the neutral and the cation); 30 unique species.

Idx	Point	Species (name, charge, spin)	Idx	Point	Species (name, charge, spin)
0	H2	H2 (0,0), H (0,1)	13	NO2	NO2 (0,1)
1	N2	N2 (0,0)	14	HN	HN (0,2), H (0,1)
2	FLi	FLi (0,0), Li (0,1)	15	O3	O3 (0,0)
3	CHN	CHN (0,0), H (0,1)	16	N2O	N2O (0,0)
4	CO2	CO2 (0,0)	17	CH3	CH3 (0,1), H (0,1)
5	F2	F2 (0,0)	18	CH2	CH2 (0,2), H (0,1)
6	C2H2	C2H2 (0,0), H (0,1)	19	H2O	H2O (0,0), H (0,1)
7	CO	CO (0,0)	20	H3N	H3N (0,0), H (0,1)
8	HLi	HLi (0,0), H (0,1), Li (0,1)	21	OH+N2 to H+N2O (BH76)	HO (0,1), N2 (0,0), H (0,1), N2O (0,0)
9	Na2	Na2 (0,0)	22	OH+CH3 to O+CH4 (BH76)	HO (0,1), CH3 (0,1), O (0,2), CH4 (0,0), H (0,1)
10	NO	NO (0,1)	23	HF+F to H+F2 (BH76)	HF (0,0), F (0,1), H (0,1), F2 (0,0)
11	CH	CH (0,1), H (0,1)	24	Li IP (IP13)	Li (0,1), Li+ (1,0)
12	HO	HO (0,1), H (0,1)	25	C IP (IP13)	C (0,2), C+ (1,1)

References: atomization energies [HK12] with H2O and C2H2 from [W411]; reaction energies [Goe17, BH76RC]; ionization potentials [NIST]. Code: dfs_pool.py, training_points.py:347-420.

2. Subsets: the descriptor distributions

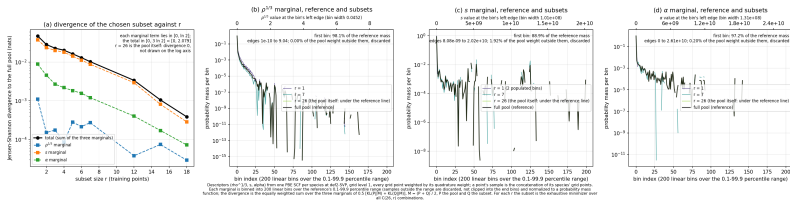
One PBE SCF per species (def2-SVP, grid level 1) [code: subset_selection.py:106-118, 448-470] gives at every grid point g with quadrature weight w_g

$$\rho^{1/3}, \quad s = \frac{|\nabla \rho|}{2(3\pi^2)^{1/3} \rho^{4/3}} \quad [\text{PBE96}], \quad \alpha = \frac{\tau - \tau_W}{\tau_{\text{unif}}} \quad [\text{SCAN15}], \text{ clipped to } [0, 100].$$

A point's sample is the concatenation of its species' grid points; the reference is the concatenation over all 26 points. Each marginal is binned into 200 equal-width bins between its 0.1 and 99.9 percentiles and normalized to a mass function:

$$h_x^{(b)} = \sum_{g: x_g \in \text{bin } b} w_g, \quad P_x^{(b)} = \frac{h_x^{(b)}}{\sum_{b'} h_x^{(b')}}.$$

Training-subset selection: Jensen-Shannon divergence to the full DFS pool



2. Subsets: the metric and the search [Lin91]

The distance between a candidate subset S and the pool is the Jensen-Shannon divergence [code: subset_selection.py:213–246, 472] summed over the three marginals with equal weights, P_x the pool and Q_x the subset mass function, probabilities floored at 10^{-12} , natural logarithm:

$$\text{JSD}(S) = \sum_{x \in \{\rho^{1/3}, s, \alpha\}} \frac{1}{2} [\text{KL}(P_x \parallel M_x) + \text{KL}(Q_x \parallel M_x)], \quad M_x = \frac{1}{2}(P_x + Q_x),$$

$$\text{KL}(P \parallel Q) = \sum_b P^{(b)} \ln \frac{P^{(b)}}{Q^{(b)}}, \quad \text{each marginal term in } [0, \ln 2].$$

For each $r \in \{1, \dots, 7, 12, 15, 18, 26\}$ the subset is the exhaustive minimizer over all $\binom{26}{r}$ combinations (9.7 million at $r = 12$); a subset with no grid point inside a marginal's range gets an infinite divergence.

- The linear edges reach the low-density tails (s to 2.0×10^{10} , α to 2.6×10^{10} , $\rho^{1/3}$ from 10^{-10} to 9.04), so the first bin holds 98.1, 88.9 and 97.2 percent of the weighted reference mass: the whole chemical range for s and α , the vacuum tail ($\rho < 9.2 \times 10^{-5}$) for $\rho^{1/3}$.
- The metric therefore compares the low-density tails of the molecular grids (s , α) and the valence and core region ($\rho^{1/3}$); the s term is 74 to 89 percent of every divergence.

2. Subsets: the chosen subsets [ledger_subset_index_log.json; plot_subset_jsd.py]

<i>r</i>	Points (ledger index order)	AE / BH76 / IP	JSD	$\rho^{1/3}$	<i>s</i>	α
1	H2O	1 / 0 / 0	0.0475	0.00109	0.03764	0.00881
2	HLi, OH+N2_to_H+N2O	1 / 1 / 0	0.0284	0.00015	0.02372	0.00455
3	OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP	0 / 2 / 1	0.0228	0.00017	0.01995	0.00267
4	CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP	1 / 2 / 1	0.0203	0.00008	0.01801	0.00218
5	HLi, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP, C_IP	1 / 2 / 2	0.0163	0.00028	0.01418	0.00183
6	HLi, CH, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, Li_IP, C_IP	2 / 2 / 2	0.0130	0.00021	0.01120	0.00156
7	CHN, CO2, HLi, CH, OH+N2_to_H+N2O, Li_IP, C_IP	4 / 1 / 2	0.0102	0.00027	0.00877	0.00120
12	CO2, C2H2, CO, HLi, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	7 / 3 / 2	0.0034	0.00004	0.00292	0.00040
15	FLi, CHN, CO2, C2H2, CO, HLi, NO, CH, NO2, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	10 / 3 / 2	0.0010	0.00007	0.00081	0.00017
18	N2, FLi, CHN, CO2, C2H2, CO, HLi, Na2, NO, CH, NO2, N2O, CH2, OH+N2_to_H+N2O, OH+CH3_to_O+CH4, HF+F_to_H+F2, Li_IP, C_IP	13 / 3 / 2	0.0004	0.00003	0.00028	0.00007
26	the whole pool	21 / 3 / 2	0	0	0	0

- The $\rho^{1/3}$, *s* and α columns are the three marginal terms of the JSD (their sum is the total); every value is recomputed from the descriptor caches and agrees with the ledger to 10^{-9} . The subsets are nested only where the minimizer happens to nest; the 2-to-6-point subsets are barriers and ionizations with one or two molecules.

3. Pre-training: targets and objective

Each network is fitted pointwise [code: pretrain.py:261-341, pretrain_data_gen.py] to reproduce its parent (PBE on the GGA rung, SCAN on the meta-GGA) on the parent's own densities of the DFS pre-training inventory plus every atom of the BH76 and W4-11 pools (39 systems GGA, 38 meta-GGA), with a synthetic (r_s, s, α) mesh [DFS21 SI] at 30 percent of the weight:

$$t_x = F_x^{\text{parent}} - 1 = \frac{\epsilon_x^{\text{parent}}}{\epsilon_x^{\text{LDA}}} - 1, \quad t_c = F_c^{\text{parent}} - 1 = \frac{\epsilon_c^{\text{parent}}}{\epsilon_c^{\text{PW92}}} - 1,$$

the exchange rows posed per spin channel at the doubled density $(2n_\sigma, 4\sigma_{\sigma\sigma})$ [OP79]. The objective is the integration-weighted pointwise residual plus a per-system energy term:

$$\mathcal{L}_{\text{pre}} = \frac{\sum_i w_i (F_i^{\text{NN}} - 1 - t_i)^2}{\sum_i w_i} + w_E \frac{1}{N_{\text{sys}}} \sum_s \left(\sum_{i \in s} w_i^{\text{grid}} n_i \epsilon_i^{\text{LDA}} F_i^{\text{NN}} - E_{xc,s}^{\text{parent}} \right)^2,$$
$$w_i = |n_i \epsilon_i^{\text{LDA}}| w_i^{\text{grid}}, \quad w_E = 0.1.$$

- The energy term is needed because the pointwise residual alone does not bound a system's energy.
- Adam, 20000 steps, learning rate 10^{-3} to 10^{-5} from the midpoint, clip 1.0, a 20 percent validation slice scored every 50 steps with patience 300 checks [cfg pretrain:].

3. Pre-training: the fidelity certificate

The saved networks are re-evaluated [code: cluster/fidelity.py, grid_config.py:287-312] on every system at the production basis and grid and compared with the parent:

$$\Delta E_{xc,s} = E_{xc,s}^{\text{NN}} - E_{xc,s}^{\text{parent}}, \quad \Delta AE = \Delta E_{xc,\text{mol}} - \sum_A \Delta E_{xc,A},$$

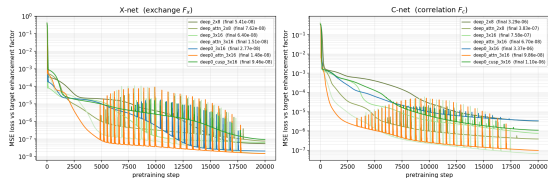
$$\max_{\text{atoms}} |\Delta E_{xc,A}| \leq 1.0 \text{ mHa}, \quad \text{mean}_{\text{mol}} |\Delta AE| \leq 1.0 \text{ kcal/mol}, \quad \max_{\text{mol}} |\Delta AE| \leq 2.0 \text{ kcal/mol}.$$

A failing certificate holds the group's training array; every trained cell starts from a network that reproduces its parent to within these tolerances.

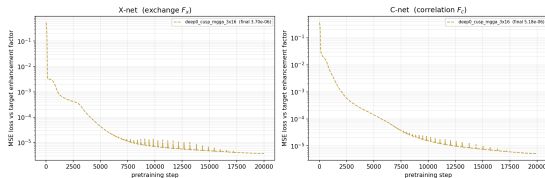
Architecture	Parent	Loss X / C	max atom (mHa)	mean $ \Delta AE $	max $ \Delta AE $	Verdict
deep_3x16	PBE	6.4e-8 / 7.6e-7	0.210	0.162	0.519	PASS
deep_attn_3x16	PBE	1.5e-8 / 6.7e-8	0.058	0.031	0.115	PASS
deep0_3x16	PBE	2.8e-8 / 3.4e-6	0.439	0.114	0.450	PASS
deep0_attn_3x16	PBE	1.5e-8 / 9.9e-8	0.123	0.042	0.149	PASS
deep0_cusp_3x16	PBE	9.5e-8 / 1.1e-6	0.439	0.188	1.071 (AlCl3)	PASS
deep0_cusp_mgga_3x16	SCAN	3.7e-6 / 5.2e-6	3.132	2.062	7.257 (14 species)	FAIL

3. Pre-training: the loss curves

Pretraining loss curves -- 7 archs, 20000 steps
run_20260902T145247Z



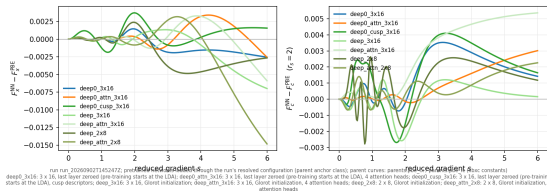
Pretraining loss curves -- 1 archs, 20000 steps
run_20260902T145250Z



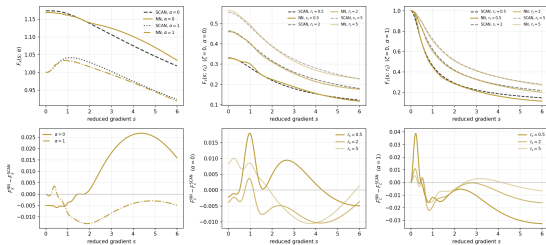
- Pointwise residual per step, exchange (left) and correlation (right); the spikes from step 5000 are the validation snapshot reloads.
- GGA clones end at 1.5×10^{-8} to 9.5×10^{-8} (exchange) and 6.7×10^{-8} to 3.4×10^{-6} (correlation); deep_attn_3x16 descends earliest and lowest, at the longest fit (28 h against 3.6 h for its plain twin).
- The SCAN clone (right) stops fifty to a hundred times higher after the same 20000 steps and fails the certificate (previous slide). Its four siblings are pending; the meta-GGA training array is held.

3. Pre-training: the pre-trained networks against their parents

Pre-trained networks against the PBE parent: differences, all architectures



deep0_cusp_mgga_3x16: pre-trained network against the SCAN parent



- Left: the difference $F^{\text{NN}} - F^{\text{PBE}}$ of every GGA clone, exchange at $n = 1$ and correlation at $r_s = 2$, along s . Within 10^{-3} for $s < 2.5$, at most 0.007 at $s = 6$ (deep_3x16 exchange); the three zero-init clones within 4.2×10^{-3} everywhere.
- Right: the SCAN clone. $F_x(\alpha = 0)$ is 0.01 to 0.027 above SCAN for $s > 2.5$, $F_c(\alpha = 1)$ 0.005 to 0.033 below it from $s = 2$; same-sign offsets over wide s ranges do not cancel between a molecule and its atoms.

4. In-sample: the training objective [DFS21 Eqs. 15-17]

One optimizer step per training group per epoch (a reaction, or a molecule with its density and potential references); the group loss is the DFS weighting ($\lambda_{RE} = 1$, $\lambda_n = 20$), with E^{NN} the self-consistent energy of the $N = 3$ cycle SCF and t the cycle:

$$\mathcal{L} = 1 \cdot \mathcal{L}_{AE} + 1 \cdot \mathcal{L}_{\text{rxn}} + 1 \cdot \mathcal{L}_{IP} + 1 \cdot \mathcal{L}_{V_{xc}} + 20 \cdot \mathcal{L}_{\rho},$$

$$\mathcal{L}_{\text{rxn}} = \frac{1}{|T|} \sum_{t \in T} w_t^2 r_t^2, \quad r_t = \sum_i c_i E_{i,t}^{\text{NN}} - E^{\text{ref}}, \quad w_t = \left(\frac{t}{N-1} \right)^2,$$

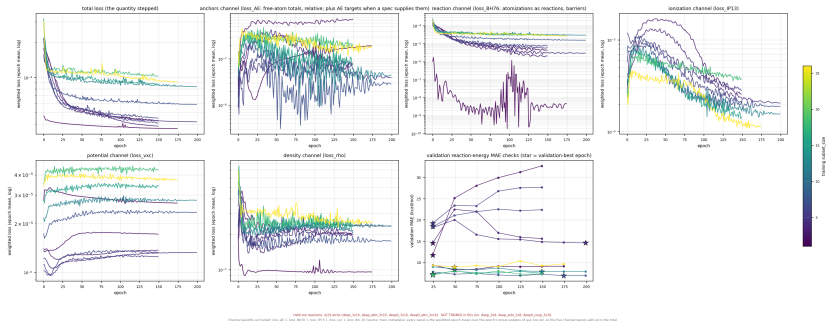
$$\mathcal{L}_{AE} = 0.01 \text{mean}_Z \frac{(E_Z^{\text{NN}} - E_Z^{\text{anchor}})^2}{(E_Z^{\text{anchor}})^2}, \quad \mathcal{L}_{IP} = \text{mean} (E_+^{\text{NN}} - E_0^{\text{NN}} - IP^{\text{ref}})^2, \quad \mathcal{L}_{V_{xc}} = \frac{\|V_{xc}^{\text{NN}} - V_{xc}^{\text{ref}}\|_F^2}{n_{\text{AO}}^2},$$

$$\mathcal{L}_{\rho} = \text{mean}_{\text{mol}} \frac{\sum_g w_g (n_g^{\text{NN}} - n_g^{\text{ref}})^2}{N_e^2}, \quad N_e = \sum_g w_g n_g^{\text{ref}}.$$

- $\mathcal{L}_{\text{rxn}}$ is the DFS convergence-tail scoring ($w_t^2 = 0, 1/16, 1$ at $N = 3$) over the W4-11 atomizations, as reactions with the network's own atoms, and the BH76 barriers [W411, GMTKN55]; the free atoms are held near their exact energies [Chak93]; V_{xc} is compared on the reference density; the density against CCSD on the same grid, per electron. Code: losses.py:76, 366-396; train.py:1829-1835; oneshot.py:632-653.

4. In-sample: the training losses (deep_3x16)

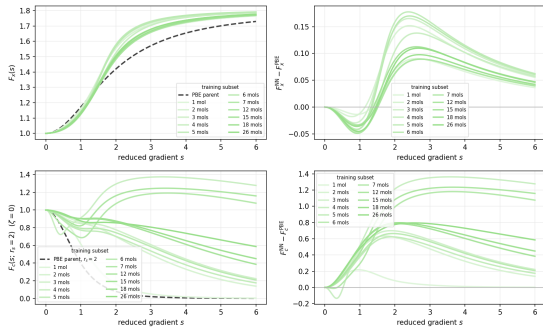
Per-channel training losses -- deep_3x16 (weighted epoch means; panels sum to the total) · run_20260902T145247Z · val-best



- SCF: 3 cycles from the converged PBE density, DFS mixing $D_{t+1} = (1 - a_t)D_t + a_t D_t^{\text{out}}$, $a_t = 0.3^t + 0.3$ [DFS21 SI; code: solver.py:305-316], gradients through every cycle, the last mixed density scored. AdamW $10^{-3} \rightarrow 10^{-5}$ over the second half, weight decay 10^{-4} , clip 1.0, 200 epochs, validation MAE on 47 reactions every 25 epochs [cfg hyperparams:].
- The reaction channel falls in every cell, by 3x (deep0_3x16 at 18) to 190x (deep_attn_3x16 at 2); the density channel by 1.3x to 4.6x, both settled within 25 epochs. The potential channel, absent from the DFS objective, rises from seven points upward (deep_3x16 at 18: $3.9\text{e-}5$ to $4.3\text{e-}5$).
- Validation MAE: the 2-to-6-point cells of deep_3x16 and deep0_3x16 rise after the first check (deep_3x16 at 2: 11.7 to 32.7 kcal/mol, best at epoch 25); the cells from seven points hold 6 to 9 kcal/mol; deep_attn_3x16's small cells fall (37 to 7 kcal/mol).

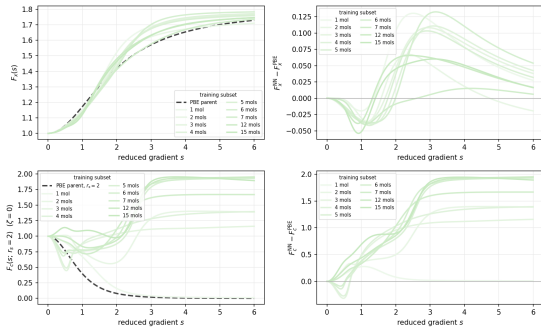
4. In-sample: the trained enhancement factors

deep_3x16: trained networks against the PBE parent



run run_20200902T145247Z; checkpoint channel val_best (model_val_best.exp); slices: F_x at r_s=1; zero extra descriptors: F_x at r_s=0, r_s=2; 11 completed cell(s); max(|F_x|) 1.78e-01; parent curves parents.pbe_3x / parents.pbe_3x (libxc constants); deep_3x16: 3 x 16, Glorot initialization

deep_attn_3x16: trained networks against the PBE parent



run run_20200902T145247Z; checkpoint channel val_best (model_val_best.exp); slices: F_x at r_s=1; zero extra descriptors: F_x at r_s=0, r_s=2; 9 completed cell(s); max(|F_x|) 1.32e-01; parent curves parents.pbe_3x / parents.pbe_3x (libxc constants); deep_attn_3x16: 3 x 16, Glorot initialization, 4 attention heads

- Exchange, every architecture and subset: the trained F_x is below PBE by 0.02 to 0.06 near $s = 0.8$ and above it from $s \approx 1.4$, by up to 0.18 (deep_3x16), 0.13 (deep_attn_3x16), 0.14 (deep0_3x16) at $s = 2.5$ to 3; the one-molecule cell moves least.
- Correlation at $r_s = 2$: between 0.6 and 1.0 up to $s = 2$ (PBE falls to 0.02 by $s = 3$), then cell-dependent: deep_3x16 at 7 to 26 points decays to 0.4 to 0.6, at 2 and 3 points rises to 1.1 to 1.4; every deep_attn_3x16 cell but the one-molecule one rises above 1 (up to 1.9 at $s = 6$).

5. Held-out: the set and the metrics [Goe17; DFS21 Eqs. 20, 21]

Test set: GMTKN55 BH76 barriers and W4-11 atomization energies, minus the 47-reaction validation slice, each cell's training reactions and four duplicate barriers: 58 BH76 and 120 W4-11 reactions, 189 species with CCSD densities. PBE is evaluated on each cell's own slice; the functionals with the same 3-cycle SCF they trained with, PBE's density converged.

$$\text{WTMAD-2} = \frac{56.84}{N} \sum_{i \in \{\text{BH76}, \text{W4-11}\}} N_i \frac{\text{MAD}_i}{|\Delta E_{\text{ref}}|_i}, \quad \varepsilon_n = \frac{1}{N_e} \int |n - n^{\text{ref}}| d^3r, \quad N_e = \int n^{\text{ref}} d^3r,$$

$$\mathcal{ED} = \frac{2}{1/E + 1/(\gamma D)}, \quad D_{\text{RMSE}} = \sqrt{\sum_g w_g (n_g - n_g^{\text{ref}})^2 / \sum_g w_g}.$$

- WTMAD-2 over the two present subsets; alone, a pool reads $56.84 \text{ MAD}_i / |\Delta E_{\text{ref}}|_i$, a scaled relative error. $\gamma = 1084.87 \text{ kcal/mol}$ on ε_n (DFS units) or $\gamma = E_{\text{PBE}}/D_{\text{PBE}}$ on the grid RMSE (self-calibrated).
- A cell beats PBE when its $\mathcal{ED}$ is below PBE's on the same slice. Pooled PBE: 8.92 kcal/mol WTMAD-2, ε_n 0.00923.
- The harmonic mean is set by its smaller leg: with $\gamma \varepsilon_n \approx 10 \text{ kcal/mol}$ here, 1 kcal/mol of E moves $\mathcal{ED}$ by 0.4 to 0.7, the whole density spread by at most 1.5. Code: `evaluation.py:217`; `make_ablation_arch_figure.py:3715-3750, 4840`.

5. Held-out: per-cell results, combined leg, DFS units [holdout_by_pool_3x3_eps.csv]

Arch	r	E NN	E PBE	eps NN	eps PBE	ED NN	ED PBE	Δ ED	Arch	r	E NN	E PBE	eps NN	eps PBE	ED NN	ED PBE	Δ ED
deep_3x16	1	7.92	8.92	0.00951	0.00923	8.96	9.46	-0.50	deep_attn_3x16	7	5.94	8.92	0.00963	0.00923	7.57	9.49	-1.91
deep_3x16	2	9.61	8.92	0.01071	0.00923	10.52	9.44	+1.08	deep_attn_3x16	12	6.61	8.92	0.01019	0.00923	8.27	9.56	-1.29
deep_3x16	3	10.67	8.92	0.01068	0.00923	11.11	9.32	+1.79	deep_attn_3x16	15	6.47	8.92	0.00956	0.00923	7.97	9.60	-1.63
deep_3x16	4	11.06	8.92	0.01182	0.00923	11.88	9.34	+2.53	deep0_3x16	1	8.18	8.92	0.00997	0.00923	9.32	9.46	-0.14
deep_3x16	5	13.97	8.92	0.01113	0.00923	12.96	9.32	+3.64	deep0_3x16	2	9.16	8.92	0.01143	0.00923	10.54	9.44	+1.10
deep_3x16	6	13.51	8.92	0.01091	0.00923	12.62	9.34	+3.28	deep0_3x16	3	7.99	8.92	0.01062	0.00923	9.44	9.32	+0.12
deep_3x16	7	6.44	8.92	0.01017	0.00923	8.13	9.49	-1.35	deep0_3x16	4	8.06	8.92	0.01160	0.00923	9.83	9.34	+0.48
deep_3x16	12	7.41	8.92	0.01047	0.00923	8.97	9.56	-0.59	deep0_3x16	5	10.48	8.92	0.01099	0.00923	11.16	9.32	+1.83
deep_3x16	15	6.89	8.92	0.01004	0.00923	8.44	9.60	-1.16	deep0_3x16	6	10.40	8.92	0.01097	0.00923	11.10	9.34	+1.76
deep_3x16	18	6.74	8.92	0.00999	0.00923	8.31	9.62	-1.31	deep0_3x16	7	5.90	8.92	0.01051	0.00923	7.77	9.49	-1.72
deep_3x16	26	8.23	8.92	0.00973	0.00923	9.25	9.80	-0.55	deep0_3x16	12	7.11	8.92	0.01058	0.00923	8.78	9.56	-0.78
deep_attn_3x16	1	8.26	8.92	0.00925	0.00923	9.06	9.46	-0.40	deep0_3x16	15	7.10	8.92	0.01013	0.00923	8.63	9.60	-0.97
deep_attn_3x16	2	7.87	8.92	0.00972	0.00923	9.01	9.44	-0.43	deep0_3x16	18	6.82	8.92	0.00996	0.00923	8.37	9.62	-1.26
deep_attn_3x16	3	7.22	8.92	0.00981	0.00923	8.60	9.32	-0.72	deep0_3x16	26	8.52	8.92	0.00961	0.00923	9.38	9.80	-0.42
deep_attn_3x16	4	6.09	8.92	0.01020	0.00923	7.85	9.34	-1.49	deep0_attn_3x16	1	8.11	8.92	0.00906	0.00923	8.89	9.46	-0.57
deep_attn_3x16	5	6.03	8.92	0.00883	0.00923	7.40	9.32	-1.92	deep0_attn_3x16	2	13.64	8.92	0.01168	0.00923	13.14	9.44	+3.70
deep_attn_3x16	6	6.09	8.92	0.00973	0.00923	7.73	9.34	-1.61									

- E in kcal/mol (2-subset WTMAD-2), eps per electron, ED under $\gamma = 1084.87$ kcal/mol; E PBE and eps PBE are the pooled values, ED PBE the cell-slice value. Δ ED = ED NN – ED PBE: **bold, negative** marks a cell below PBE (22 of 33).
- Verdicts within about 5 percent of PBE depend on the leg and the units: under the plain-MAE leg deep_attn_3x16 at 3, 5 and 6 flip; under self-calibrated grid-RMSE units deep_3x16 at 2, deep_attn_3x16 at 2 and deep0_3x16 at 2 and 3 flip (holdout_ed_combined.csv). The deep0_attn_3x16 verdicts are the same under every choice.

6. Pending work

- Cells still queued: deep_attn_3x16 at 12 to 26, deep0_attn_3x16 and deep0_cusp_3x16 at every subset.
- Meta-GGA group: four certificates pending; the deep0_cusp_mgga_3x16 FAIL holds the training array. The choice between dropping the architecture, refitting with more capacity or a higher energy-term weight, and an anchored form follows the four results.
- Converged-SCF re-evaluation of the 28 finished cells (DIIS at 10^{-8} Ha) and the CCSD T1 diagnostics for a model-free multireference list: two cluster jobs pending.
- Two training arms on the deep_3x16 architecture at 7, 12, 15, 18 and 26 points: 25 SCF cycles with the current protocol (arm A) and the dpyscf-parity protocol (arm B: no convergence freeze, the executed first-mix index, the randomized atomic/PBE seed, Adam with the plateau schedule, the DFS L2 and channel weights). Submitted 2026-09-08, no results yet.
- The figure suite's in-sample atomization-energy panel stays out of the documents until it uses the network's own atoms (it now subtracts fixed exact atomic energies and reports the absolute-energy offset).

References and sources

- DFS21 S. Dick and M. Fernandez-Serra, Phys. Rev. B 104, L161109 (2021), with its Supplemental Material; reference code dpyscf.
- PBE96 J. P. Perdew, K. Burke and M. Ernzerhof, Phys. Rev. Lett. 77, 3865 (1996). [PW92] J. P. Perdew and Y. Wang, Phys. Rev. B 45, 13244 (1992).
- SCAN15 J. Sun, A. Ruzsinszky and J. P. Perdew, Phys. Rev. Lett. 115, 036402 (2015). [PRSB14] J. P. Perdew, A. Ruzsinszky, J. Sun and K. Burke, J. Chem. Phys. 140, 18A533 (2014).
- OP79 G. L. Oliver and J. P. Perdew, Phys. Rev. A 20, 397 (1979). [Kato57] T. Kato, Commun. Pure Appl. Math. 10, 151 (1957). [Steiner63] E. Steiner, J. Chem. Phys. 39, 2365 (1963).
- Vas17 A. Vaswani et al., NeurIPS 30 (2017). [Xio20] R. Xiong et al., ICML (2020), arXiv:2002.04745.
- Lin91 J. Lin, IEEE Trans. Inf. Theory 37, 145 (1991).
- Goe17 L. Goerigk et al., Phys. Chem. Chem. Phys. 19, 32184 (2017) (GMTKN55, WTMAD-2, BH76RC). [W411] A. Karton, S. Daon and J. M. L. Martin, Chem. Phys. Lett. 510, 165 (2011). [HK12] R. Haunschild and W. Klopper, J. Chem. Phys. 136, 164102 (2012). [Chak93] S. J. Chakravorty et al., Phys. Rev. A 47, 3649 (1993). [NIST] NIST Chemistry WebBook ionization energies.
- code the repository: xcquinox/alec/networks.py, descriptors.py, metagga.py, config.py, xcquinox/net.py, xcquinox/alec/subset_selection.py, dfs_pool.py, training_points.py, pretrain.py, cluster/fidelity.py, cluster/grid_config.py, losses.py, train.py, oneshot.py, solver.py, evaluation.py, notebooks/analysis/make_ablation_arch_figure.py; the figures and CSVs under notebooks/analysis/figures_dfs_step7_*; [cfg] the run configurations hpcjobs/configs/dfs_step7.dfs6311_grid3_v7*.yaml.